

Daniele Famà

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EDUCATION

Télécom Paris, Institut Polytechnique de Paris

2025 – 2027

M.Sc. in Engineering Double Degree Program

Paris, France

- Applied Algebra (Cryptography, Quantum Information, Coding Theory)
- Distributed Software Systems, Data Science

Politecnico di Torino

2024 – 2025

M.Sc. in Data Science and Engineering Double Degree Program

Turin, Italy

- Focus: Machine Learning, Deep Learning, and Big Data Systems.

University of Catania

2021 – 2024

B.Sc. in Computer Engineering 110/110, summa cum laude

Catania, Italy

- Thesis: Design of a Third-Order Tow-Thomas Butterworth Filter for Audio-Band Applications

EXPERIENCE

BANERA: Energy-Aware Neural Architecture Search

2026

Green AI Research

Télécom Paris

- Developing energy-based regularization for Neural Architecture Search.
- **Fully automated** end-to-end energy measurements on Raspberry Pi and NVIDIA Jetson using INA226EVM.
- Designed adaptive workload calibration and reproducible multi-host experiments.

PROJECTS

ARDA VSLNet: Efficient Video-Language Localization on Ego4D

2025

Computer Vision & Deep Learning

Politecnico di Torino

- Enhanced VSLNet with FiLM conditioning for natural-language temporal localization in egocentric videos.
- Reduced model size by 88.47% through knowledge distillation, outperforming a similarly sized baseline by 19.73%.
- Applied block-wise distillation to cut parameters by 47.62% and GFLOPs by 80.6% while retaining near-teacher accuracy.

Free Spoken Digit Dataset Regression Problem

2025

Machine Learning

Politecnico di Torino

- Converted audio signals into spectrogram representations for classification.
- Ranked 1st on the class public leaderboard over more than 100 teams.

HACKATHON

STMicroelectronics

2023

ST Spring Robotics Championship - 1st edition

Catania, Italy

- Worked with robots without remote sensing tools to stay in a certain area
- Awarded 2nd place among participating teams.

SIDE PROJECTS

Handwritten Digit Recognition on Casio fx-9860GIII

2026

Embedded ML

- Built an INT8 neural network for on-device handwritten digit recognition.
- Achieved 95.47% MNIST accuracy with fixed-point C inference.

Singular Value Decomposition (SVD) Image Compression

2025

Matlab

- Developed an image compression algorithm leveraging Singular Value Decomposition (SVD).
- Implemented SVD from scratch.

SKILLS

Programming : Python, Java, C/C++, Matlab, PHP, JavaScript

Python libraries : Numpy, Pandas, Matplotlib, Sklearn, Pytorch, Keras

Frameworks : Hadoop, Spark, Laravel, TensorFlow

Languages : *Italian (native proficiency), English (professional working proficiency), French (elementary proficiency)*